Severity and Case Classification

Purpose and assessment approach

The assessments distinguish causality from seriousness and expectedness: a well-known, expected reaction can still be serious, and a serious outcome does not by itself prove causality.

Seriousness follows standard pharmacovigilance outcome criteria, including death, life-threatening event, inpatient hospitalization or prolonged hospitalization, persistent/significant disability, congenital anomaly, or another medically important event. Expectedness is assessed against recognized product safety information; formal regulatory listedness must be checked against the applicable local product information and version in force.

Summary of findings

Case	Suspected reaction	Causality assessment	Seriousness	Expectedness	Clinical impact
1. Enalapril	Persistent dry cough	Naranjo 6: probable; WHO–UMC: probable/likely	Non-serious in the vignette	Expected / labeled	Moderate symptom burden; resolves after withdrawal; no organ injury or admission
2. Simvastatin + clarithromycin	Rhabdomyolysis with acute kidney injury and hyperkalemia	Interaction-level WHO–UMC: probable/likely; Naranjo component scores are limited (see below)	Serious: inpatient admission; medically important acute kidney injury	Expected / known interaction	Severe, acute, reversible with hospitalization and treatment; potentially life-threatening if untreated

Important interpretation

The rhabdomyolysis case is a preventable, high-impact interaction signal for medication-use systems: clarithromycin is a strong CYP3A4 inhibitor and coadministration with simvastatin is contraindicated in current US labeling. The case also reports simvastatin 40 mg/day with amlodipine, above the 20 mg/day maximum cited in the source case. The interaction provides a strong pharmacologic explanation, while attribution to either component drug alone is less certain.

Case 1 Enalapril induced dry cough

Complete case narrative

A 56-year-old woman with hypertension begins enalapril 5 mg orally once daily. Her other long-term medicine is amlodipine 5 mg once daily. She has no known asthma, chronic lung disease, smoking history, or prior ACE inhibitor intolerance. About three weeks after starting enalapril, she develops a persistent, dry, tickling cough, especially at night. She has no fever, sputum, wheezing, dyspnea, facial or tongue swelling, or symptoms suggesting an acute respiratory infection. Examination and oxygen saturation are normal; chest radiograph is unremarkable. No laboratory abnormality or hospitalization is reported. Enalapril is discontinued and replaced with losartan. The cough resolves over the next two weeks, with no recurrence during follow-up. Rechallenge is not attempted.

Suspected medicine and reaction

Item	Assessment
Suspected drug	Enalapril, 5 mg once daily; started approximately three weeks before cough onset.
Reaction	Persistent nonproductive cough; consistent with a recognized ACE inhibitor adverse effect.
Mechanistic plausibility	ACE inhibition reduces bradykinin degradation; accumulation of bradykinin and related mediators is implicated in persistent dry cough.
Dechallenge	Cough resolved after enalapril withdrawal; switch to losartan provides a plausible alternative antihypertensive.
Alternative causes	Infection, asthma, smoking-related disease, and radiographic abnormality are not supported by the illustrative history.

Naranjo assessment

Question	Response	Score	Reason
Previous conclusive reports?	Yes	+1	ACE inhibitor associated persistent dry cough is recognized.
Event after drug given?	Yes	+2	Cough began about three weeks after enalapril initiation.
Improved on withdrawal / antagonist?	Yes	+1	Resolved after withdrawal and substitution.
Reappeared on readministration?	Unknown	0	No rechallenge.
Alternative causes could explain it?	No	+2	The vignette reports no credible alternative cause after evaluation.
Reappeared with placebo?	Unknown	0	No placebo challenge.
Toxic drug concentration detected?	Unknown	0	Not measured; not clinically indicated for this event.
Dose response demonstrated?	Unknown	0	No dose increase/decrease comparison.
Similar reaction previously?	Unknown	0	No prior ACE inhibitor exposure stated.
Objective confirmation?	No	0	Cough is clinically reported; no diagnostic objective test confirms drug causation.

Total		6	Probable (score 5–8).

The score supports probable causality, but Naranjo is an aid rather than proof. Dechallenge and the known reaction pattern carry most of the support; there is no rechallenge.

WHO–UMC classification

Probable / likely. The onset has a reasonable temporal relationship; the cough is a recognized ACE inhibitor effect; improvement after withdrawal is clinically reasonable.

Seriousness and expectedness

Dimension	Classification	Rationale
Seriousness	Non-serious	No death, life-threatening episode, hospitalization, disability, or other medically important intervention is described. Persistent cough is troublesome but does not meet seriousness criteria in this scenario.
Expectedness	Expected / labeled	Persistent nonproductive cough is described in enalapril product information. For formal listedness, compare the exact product and current local reference safety information.

Clinical impact and conclusion

Clinical impact is moderate symptom burden: sleep disruption and discomfort are plausible, but there is no evidence of pulmonary injury or acute threat. Symptoms resolve after stopping enalapril. The case supports an expected, non-serious, probable ADR. In practice, assess competing causes and red flags such as angioedema or respiratory compromise, then review antihypertensive alternatives with the treating clinician.

Case 2 Simvastatin clarithromycin associated rhabdomyolysis

Complete case narrative

A 68-year-old woman weighing 58 kg with hypertension and dyslipidemia had a baseline serum creatinine of 0.9 mg/dL and no previous muscle-related complaints. She had taken simvastatin 40 mg at bedtime and amlodipine 5 mg once daily for three years, and hydrochlorothiazide 12.5 mg once daily for two years. Clarithromycin 500 mg twice daily was newly prescribed for seven days for a lower respiratory tract infection. The prescriber was unaware that she was taking simvastatin.

Day	Clinical course
0	Clarithromycin started.
3	Mild thigh and calf pain developed; initially attributed to infection.
4	Marked proximal muscle weakness, difficulty climbing stairs, and cola-coloured urine.
6	Hospital admission. Clarithromycin and simvastatin discontinued.
6–10	Intravenous fluids and electrolyte monitoring.
10	Creatine kinase (CK) began trending down and urine colour cleared.
14	Creatinine improved to 1.0 mg/dL and muscle weakness resolved.

Admission findings on Day 6

Parameter	Result	Reference range	Interpretation
Creatine kinase	18,500 U/L	<170 U/L	Marked muscle injury
Serum creatinine	1.9 mg/dL	0.5–1.1 mg/dL	Acute kidney injury relative to baseline
Potassium	5.6 mmol/L	3.5–5.0 mmol/L	Hyperkalemia; consistent with muscle-cell injury
AST	420 U/L	<40 U/L	May arise from skeletal muscle injury
ALT	210 U/L	<40 U/L	May rise with muscle injury; alone does not establish primary liver injury
Urine myoglobin	Positive	Negative	Supports myoglobin release
TSH	Normal	Within normal range	Hypothyroidism less likely

Suspected drugs and mechanism

Role	Medicine	Assessment
Precipitating interacting drug	Clarithromycin	Strong CYP3A4 inhibitor; can markedly increase simvastatin exposure.
Affected / object drug	Simvastatin	CYP3A4 substrate; increased exposure raises myotoxicity risk.
Additional risk factor	Amlodipine	Source notes patient used simvastatin 40 mg/day; source identifies 20 mg/day as the recommended maximum with amlodipine.
Less likely contributor	Hydrochlorothiazide	Used for two years without prior muscle symptoms; no clear case-specific evidence makes it a primary suspect.

Mechanism: clarithromycin inhibits CYP3A4-mediated simvastatin metabolism, increasing simvastatin exposure. Muscle-cell breakdown releases CK, myoglobin, and potassium. Myoglobin can contribute to renal injury; the patient had elevated creatinine and potassium. This is a plausible pharmacokinetic interaction with a predictable, dose-related Type A reaction.

Naranjo assessment

The Naranjo scale is designed for a suspected drug–event pair, not an interaction pair. Because both medicines were started/stopped together and neither was rechallenged, separate scores cannot establish which single drug caused the event. The table scores the interaction-associated event conservatively using the patient record.

Question	Response	Score	Reason
Previous conclusive reports?	Yes	+1	The interaction is recognized; official labeling describes increased simvastatin concentrations and rhabdomyolysis risk.
Event after suspected exposure?	Yes	+2	Symptoms began three days after clarithromycin was added to chronic simvastatin.
Improved after withdrawal / antagonist?	Yes	+1	Both drugs were stopped; CK, urine colour, renal function, and weakness improved with treatment.
Reappeared on readministration?	Unknown	0	No rechallenge.
Alternative causes?	Yes	−1	Concomitant interacting medicines and infection are present; the interaction itself complicates single-drug attribution. Other major causes were excluded.
Reappeared with placebo?	Unknown	0	No placebo challenge.
Toxic drug concentration detected?	Unknown	0	No simvastatin concentration reported.
Dose response demonstrated?	Unknown	0	No dose titration observation.
Similar reaction previously?	No / not reported	0	No previous muscle complaint is reported.
Objective confirmation?	Yes	+1	CK 18,500 U/L, positive urine myoglobin, weakness, and renal/electrolyte abnormalities.
Total		4	Possible by Naranjo (1–4).

This conservative total is “possible,” and it understates the strong interaction evidence because the instrument handles interacting-drug causality poorly. It should not be interpreted as contradicting the known, clinically important clarithromycin–simvastatin interaction.

WHO–UMC classification

For the combined clarithromycin–simvastatin exposure, probable / likely is a reasonable overall category: the event followed the interaction in a reasonable time, there is a strong pharmacological mechanism and a known interaction, and clinical improvement after withdrawal is plausible. Rechallenge is not required.

For either medicine considered alone, possible is more defensible because both were withdrawn together and component-specific causality cannot be separated confidently.

Seriousness and expectedness

Dimension	Classification	Rationale
Seriousness	Serious	The reaction required inpatient hospitalization. It also caused acute kidney injury and hyperkalemia, which are medically important complications. The patient recovered; no death or persistent disability is reported.
Expectedness	Expected / known interaction	Clarithromycin labeling states concomitant simvastatin use is contraindicated because concentrations increase and myopathy, including rhabdomyolysis, may occur. Formal listedness is assessed against the applicable product information and jurisdiction.

Clinical impact and conclusion

Clinical impact was severe and acute: marked skeletal muscle breakdown, myoglobinuria, proximal weakness, acute kidney injury, and hyperkalemia required admission, intravenous fluids, and monitoring. The patient improved by Day 14 with no persistent injury reported. The episode was potentially life-threatening without timely recognition, although the supplied facts do not state that death was imminent at the time; seriousness is established by hospitalization and medically important complications.

Conclusion: this is a serious, expected, probable interaction-associated ADR with substantial but reversible patient harm. The principal preventable safety failures were co-prescription of contraindicated clarithromycin and simvastatin without an interaction check, and simvastatin 40 mg/day with amlodipine despite the source-stated dose limit. Safer prescribing requires medication reconciliation, interaction screening, selection of a non-interacting antibiotic or statin strategy when clinically appropriate, and prompt evaluation of muscle pain, weakness, dark urine, CK, renal function, and electrolytes.

Cross-case pharmacovigilance interpretation

Assessment dimension	Enalapril cough case	Rhabdomyolysis case
Causality	Probable by Naranjo 6 and WHO–UMC probable/likely in illustrative vignette.	Combined exposure probable/likely by WHO–UMC; Naranjo 4 possible, with limited ability to resolve interaction causality.
Seriousness	Non-serious on stated facts.	Serious due to inpatient admission and important renal/electrolyte complications.
Expectedness	Known and labeled effect.	Known interaction and labeled risk; exact regulatory listedness depends on local reference product information.
Clinical impact	Persistent but reversible symptom burden; no objective injury or admission.	Severe, acute, potentially life-threatening organ injury; reversible after hospital treatment.
Preventability	Often mitigated by recognizing ACE inhibitor cough and reassessing therapy.	Strongly preventable through medication reconciliation, interaction alerts, and avoiding contraindicated combination.

Guideline-based reading of these results

- Causality asks whether the medicine plausibly caused the event. Naranjo and WHO–UMC support structured clinical judgment; neither is a substitute for case quality or expert review.
- Seriousness is outcome- and intervention-based. “Severe” symptom intensity and “serious” regulatory outcome are related but distinct. The rhabdomyolysis case is serious because it required inpatient admission and involved important complications.
- Expectedness compares the nature and specificity of the reaction with the relevant reference safety information. A known or expected ADR can still be serious and reportable.
- Clinical impact considers actual harm, treatment burden, reversibility, and plausible risk if not treated. It does not convert an individual case into an incidence estimate or prove population-level risk.
- In a formal individual case safety report, document reporter and patient identifiers as permitted, suspect/concomitant medicines, dose and dates, event onset and outcome, relevant tests, medical history, treatment, seriousness criteria, and the reference information used for expectedness.